Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 10 [2023 P2]

Test Date: 07th May 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2023-P2-Model

Time: 3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. The value of $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{r^2\tan^{-1}\left(\dfrac{2r+1}{1+\left(r^2+r\right)^2}\right)+(2r+1)\tan^{-1}(r+1)^2}{\left(r^2+r\right)^2\left(\tan^{-1}r^2\right)\tan^{-1}(r+1)^2}$ is equal to

A) $-\dfrac{\pi}{4}$ **B)** $\dfrac{4}{\pi}$ **C)** $\dfrac{\pi}{2}$ **D)** $\dfrac{2}{\pi}$

2. Rational numbers a and b chosen at random among all rational numbers in the interval [0, 2) that can be written as fraction $\dfrac{n}{d}$ where n and d are integers with $1\le d\le 5$. What is the probability that

$\left(\cos(a\pi)+i\sin(b\pi)\right)^4$ is a real number?

A) $\dfrac{3}{50}$ **B)** $\dfrac{4}{25}$ **C)** $\dfrac{41}{200}$ **D)** $\dfrac{6}{25}$

3. If one number is selected from each of the three sets A = {1,2,3,4,5,6,7}, B={1,2,3,4} and C={2,3,4,5,6,7} then probability that product of the number selected from one and sum of the numbers selected from the other two is a prime is

A) $\dfrac{1}{7}$ **B)** $\dfrac{4}{7}$ **C)** $\dfrac{13}{56}$ **D)** $\dfrac{1}{14}$

4. The number of ordered triplets of sets (A, B, C) such that

a) $A,B,C\subseteq\{1,2,3,....,8\}$

b) $|A\cap B|=|B\cap C|=|C\cap A|=2$

c) $|A|=|B|=|C|=4$

(Here, |S| denotes the number of elements in the set S)

A) 45300 **B)** 45360 **C)** 42840 **D)** 5040

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. Let $y=f(x)$ satisfies the differential equation $3x^3ydx-x^4dy=\left(y^2+x^3y\right)dx$ where $x>0, y>0$ and $x\neq\frac{1}{2}$. If $y(1)=1$ then identify which of the following statement(s) is(are) correct?

A) Area of the triangle formed by tangent and normal drawn to the curve at (1, 1) with the x-axis 1sq.unit

B) If $f(x)=1+m(x-1)$ has three distinct solutions then m>1

C) If $f(x)=1+m(x-1)$ has three distinct solutions then $m\in[0,1)$

D) If $f(x)=1+m(x-1)$ has two distinct solutions then $m\in(-\infty,0)$

6. Let $f(x)=\left(\cos^2 x\right).\left(e^{\tan x}\right), x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, then which of the following options is/are true?

A) $f'''(x)=0$ has atleast one root in $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

B) $y=f'(x)$ has a point of local maxima in $x\in\left(-\frac{\pi}{4},0\right)$

C) $f''(x)=0$ has exactly one root in $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

D) $\lim\limits_{x\to\alpha}\dfrac{x^3-6x^2+11x-6}{x^2-3x+2}=-1$

where 'α' is the total number of points of maxima and minima of $y=f'(x)$

7. The value of $\sum\limits_{n=2}^{\infty}\dfrac{n^4+3n^2+10n+10}{2^n\left(n^4+4\right)}$ is less than

A) $\frac{11}{7}$ **B)** $\frac{11}{8}$ **C)** $\frac{11}{9}$ **D)** $\frac{11}{10}$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

8. Let S be the set of circles in the coordinate plane that are tangent to each of the three circles with equations $x^2+y^2=4$, $x^2+y^2=64$, and $(x-5)^2+y^2=3$. What is the sum of the areas of all circles in S is $k\pi$. Then k =

9. There exists a unique strictly increasing sequence of non-negative integers $a_1<a_2<\ldots<a_k$ such that $\frac{2^{289}+1}{2^{17}+1}=2^{a_1}+2^{a_2}+\ldots.+2^{a_k}$, then $(k-130)$ is

10. If α is a non-real 7th root of 3 and $x=\alpha+\alpha^2$ then the value of $x^7-21\left[x^2(x+1)+x^2+x+1\right]+15$ is

11. Consider $f(x)=\left||x-2|-|3x-1|+|2x+1|\right|$ and $g(x)=\left[\left|-x^2+\frac{3}{2}x+1\right|\right]$

[Note: [y] denotes greatest integer less than or equal to y.]

If m and n are the number of points where $h(x)=f(x)g(x)$ is discontinuous and non-derivable respectively then $(m+n)$ is

12. In a drawer Mr. X has 5 pairs of socks, each pair is of different color. On Monday, he selects two individual socks at random from the 10 socks in the drawer. On Tuesday, he selects 2 of the remaining 8 socks at random and on Wednesday, two of the remaining 6 socks at random. If the probability that Wednesday is the first day he selects matching socks is $\frac{m}{n}$, where m and n are relatively prime positive integers, then the value of $\frac{m+n}{11}=$

13. Let the area of given triangle ABC be Δ. Points A_1, B_1 and C_1 are the mid points of the BC, CA and AB respectively. Point A_2 is the mid point of CA_1. Lines C_1A_1 and AA_2 meet the median BB_1 at points E and D respectively. If Δ_1 be the area of the quadrilateral A_1A_2DE. If the value of $\frac{\Delta_1}{\Delta}=\frac{m}{n}$ (where m and n are coprime), then $n-3m=$

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

A_n and B_n be square matrices of order 3 which are defined as $A_n = \left[a_{ij}\right]$ and $B_n = \left[b_{ij}\right]$ where $a_{ij} = \dfrac{2i+j}{3^{2n}}$ and $b_{ij} = \dfrac{3i-j}{2^{2n}} \forall i, j, 1 \le i, j \le 3$.

If $L = \lim\limits_{n\to\infty} Tr\left(3A_1 + 3^2 A_2 + 3^3 A_3 + \ldots + 3^n A_n\right)$,

$M = \lim\limits_{n\to\infty} Tr\left(2B_1 + 2^2 B_2 + 2^3 B_3 + \ldots + 2^n B_n\right)$

(where $Tr(P)$= Trace of matrix P). Let $p, n \in N$, $f\left(x\right) = 1 - x^p$ and

$g_n\left(x\right) = \dfrac{n}{\dfrac{1}{f\left(x\right)} + \dfrac{1}{f\left(2x\right)} + \ldots + \dfrac{1}{f\left(nx\right)}}$ and $R = \lim\limits_{x\to 0} \dfrac{1 - g_n\left(x\right)}{x^p}$ at n = 5 and p = 3

14. The value of 100-R is equal to

15. The value of L + M + R is equal to

PARAGRAPH-II

ABCD is a trapezium, AB||DC, AB=8, CD=2. A semi circle with diameter AD touches side BC. Now a circle is tangent to this semicircle, side AB, and side BC, then answer the following

16. If the radius of semi circle is R then $\dfrac{23}{5}R$ equals

17. If the radius of circle is r then the value of $\dfrac{1}{5}\left(r + \sqrt{20}\right)^3$ =

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

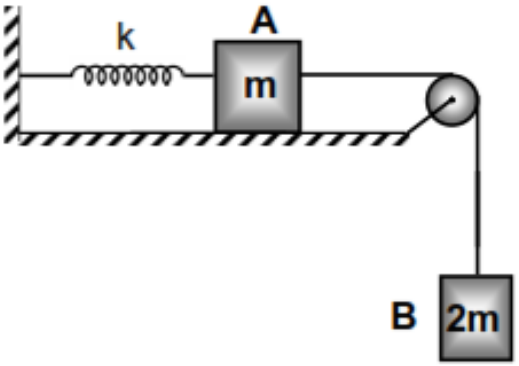
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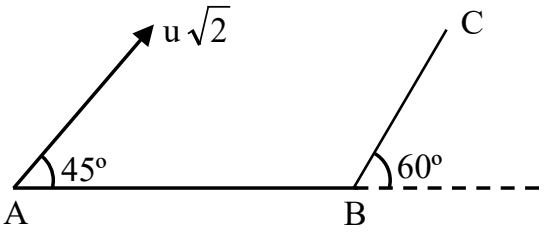
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18. In the arrangement shown, the blocks A and B are having masses m and 2m respectively and the spring constant is k. The coefficient of friction between the block A and the horizontal surface is $\mu = 0.5$. Initially the spring is relaxed and the system is released from rest. The maximum speed acquired by each block is



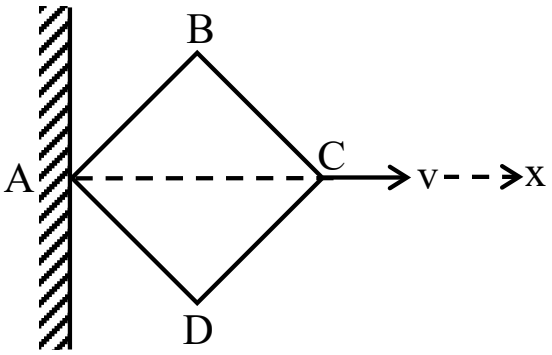
A) $g\sqrt{\frac{3m}{k}}$ **B)** $g\sqrt{\frac{3m}{2k}}$ **C)** $\frac{g}{2}\sqrt{\frac{3m}{k}}$ **D)** $\frac{g}{2}\sqrt{\frac{3m}{2k}}$

19. A particle is projected from a point A with velocity $u\sqrt{2}$ at an angle of 45º with horizontal as shown in figure. It strikes the plane BC at right angles. The velocity of the particle at the time of collision is:



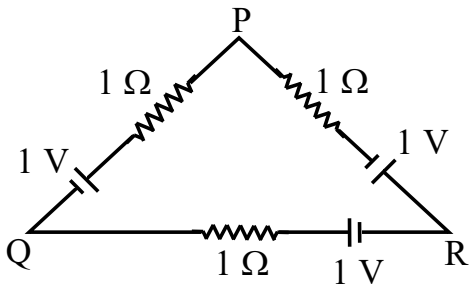
A) $\frac{\sqrt{3}u}{2}$ **B)** $\frac{u}{2}$ **C)** $\frac{2u}{\sqrt{3}}$ **D)** u

20. Four rods each of length *l* have been hinged to form a rhombus. Vertex A is fixed to rigid support, vertex C is being moved along the x-axis with a constant velocity v as shown in the figure. The rate at which vertex B is approaching the x-axis at the moment the rhombus is in the form of a square is _____



A) $\frac{v}{4}$ **B)** $\frac{v}{3}$ **C)** $\frac{v}{2}$ **D)** $\frac{v}{\sqrt{2}}$

21. Three batteries of emf 1V and internal resistance 1Ω each are connected as shown. Effective emf of combination between the points PQ is –



A) Zero **B)** 1V **C)** 2V **D)** $\frac{2}{3}$ V

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

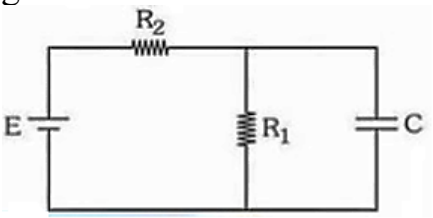
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Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
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22. For the given circuit as shown in figure $R_1 = (6.0 \pm 0.03) \times 10^1 \Omega$, $R_2 = (3.0 \pm 0.06) \times 10^1 \Omega$, $C = (1.0 \pm 0.01) F$, $E = (4.0 \pm 0.04) \times 10^1 V$. At t = 0, the circuit is formed and the capacitor is initially uncharged.



A) Maximum error in calculation of maximum current in branch of capacitor can be 0.4A

B) Maximum error in calculation of maximum current in branch of capacitor can be 0.04A

C) Maximum fractional error in calculation of time constant of circuit during charging can be 25×10^{-3}

D) Maximum fractional error in calculation of time constant of circuit during charging can be 5×10^{-3}

23. In a resonance tube experiment, a closed organ pipe of length 120 cm is used with a tuning fork of frequency 340 Hz. If water is poured in the pipe then (given velocity of sound in air = 340 m/sec.):

A) minimum length of water column to have the resonance is 45 cm.

B) the distance between two successive nodes is 50 cm.

C) the maximum length of water column to create the resonance is 95 cm

D) None of these

24. Three charged particles are in equilibrium under their electrostatic forces only –

A) The particles must be collinear
B) All the charges cannot have the same magnitude
C) All the charges cannot have the same sign.
D) The equilibrium is unstable

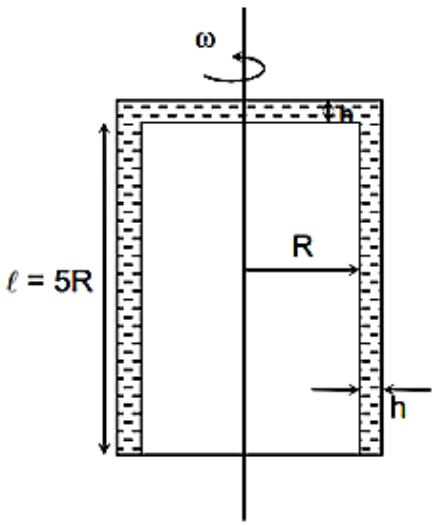
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. Nuclei of a radioactive element A are being produced at the rate of t^2 at time instant t. The element A has a decay constant λ. If N is the number of active nuclei of element A at any time t, then $\frac{dN}{dt}$ is found to be minimum at $t = 4s$. Value of N at t = 4s is $\alpha\left|\frac{(1-2\lambda)}{\lambda^2}\right|$. Find the value of α.

26. A solid cylinder of radius R and length $l = 5R$ is located coaxially in a vertical cylindrical cavity filled with a viscous liquid of viscosity η as shown in the figure. The clearance between the solid cylinder and the cavity is h (<<R). If the torque required to rotate the solid cylinder with a constant angular velocity ω about its vertical axis is $\left(\frac{K\pi\eta\omega R^4}{2h}\right)$. Find the value of K



27. A series LCR circuit containing a resistance of 120Ω has angular resonance frequency $4 \times 10^5 \text{rads}^{-1}$. At resonance the voltage across resistance and inductance are 60 V and 40 V respectively. At what frequency the current in the circuit lags the voltage by 45°. Give answer in …….. $\times 10^5\, rad/sec$.

28. When a point object is placed at a distance of 25cm from a mirror, the magnification is m_1. The object is moved 15 cm away from mirror with respect to the earlier position along principal axis, magnification becomes m_2. If $m_1 \times m_2 = 4$, the focal length of the mirror is …..… $\times 10cm$. Fill in the blank.

29. In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm, what will be number of fringes observed in the same segment of screen.

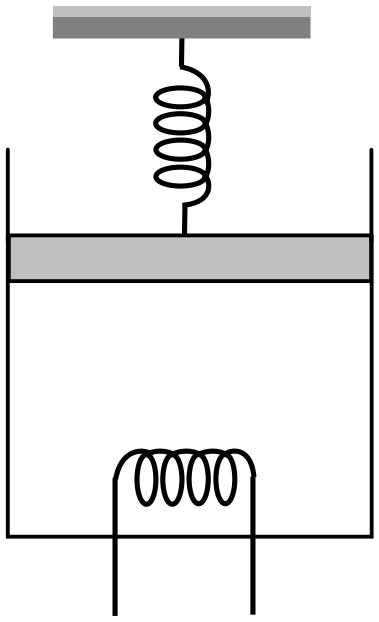
30. Find the wavelength of K_α line (in picometer) in copper (Z = 29) if the wavelength of K_α - line in iron (Z = 26) is known to be equal to193 picometer.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
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Paragraph-I

Two moles of a monoatomic ideal gas $\left[U = \frac{3}{2}nRT\right]$ is enclosed in an adiabatic, vertical cylinder fitted with a smooth, light adiabatic piston. The piston is connected to a vertical spring of spring constant 200 N/m as shown in figure. The area of cross-section of the cylinder is $20.0\ cm^2$. Initially, the spring is at its natural length and temperature of the gas is 300K. The atmospheric pressure is 100 kPa. The gas is heated slowly for some time by means of an electric heater so as to move the piston up through 10 cm. (R=8.3J/mol-K)

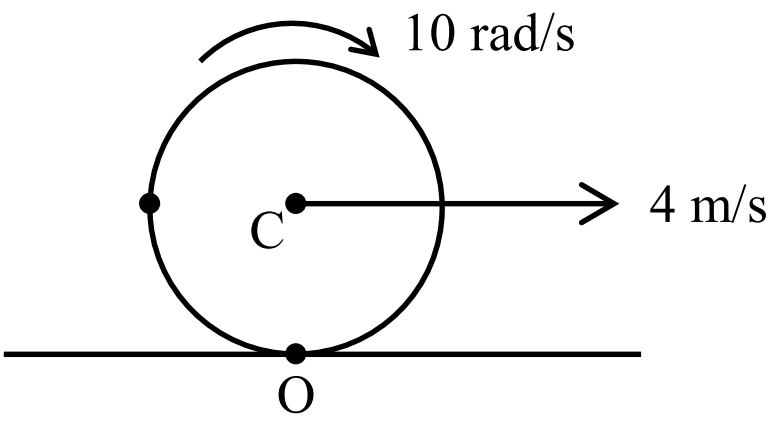


31. The work done (in Joule) by the gas in the whole process is_____

32. The quantity of heat supplied (in Joule) by the heater is ____

Paragraph-II

A disc of radius 20 cm is rolling with slipping on a flat horizontal surface. At a certain instant, the velocity of its centre is 4 m/s and its angular velocity is 10 rad/s. The lowest contact point is O.



33. Velocity (in m/s) of point O is –

34. The distance (in meter) of instantaneous center of rotation from the point O is –

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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35. Among the following options, select the options in which each complex of Set – I shows geometrical isomerism and Set II shows ionization isomerism.

1) Set I $\rightarrow [NiCl_2Br_2]^{-2}$ and $[PdCl_2Br_2]^{-2}$

Set II $\rightarrow [Pt(NH_3)_3Cl]Br$ and $[Ni(H_2O)_3Cl]Br$

2) Set I $\rightarrow [PdCl_2Br_2]$ and $[Cr(Gly)_3]^{+3}$

Set II $\rightarrow [Cr(H_2O)_5NO_2]Cl$ and $[Pt(H_2O)_3NO_2]Cl$

3) Set I $\rightarrow [Cr(H_2O)_4Cl_2]Br$ and $[Cr(ox)_3]$

Set II $\rightarrow [Zn(H_2O)_3Cl]Br$ and $[Ni(NH_3)_3Cl]Br$

4) Set I $\rightarrow [Fe(H_2O)_4Cl_2]$ and $[Pt(Gly)_2]$

Set II $\rightarrow [Cr(NH_3)_5SO_4]Cl$ and $[Co(NH_3)_5Cl]Br$

Where ox = oxalate

Gly = Glycinato

A) 1 & 4 only **B)** 1, 2 & 4 **C)** 2, 3 & 4 **D)** 2 & 4 only

36. $C_2H_6 \xrightarrow[O_2/\Delta]{Mo_2O_3} X \xrightarrow[conc\ NaOH]{HCHO(excess)} A + B$

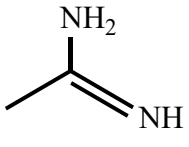
$A \xrightarrow{H^+} C \xrightarrow{Ag^+} Ag\downarrow$

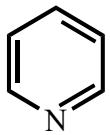
$B \xrightarrow[dry\ HCl]{excess\ 'X'} Y$

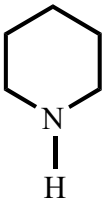
Number of stereoisomers and oxygen atoms in Y _____ respectively

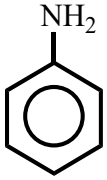
A) 2, 4 **B)** 4, 4 **C)** 8, 4 **D)** 2, 2

37. Correct basic strength order of following is

a) 

b) 

c) 

d) 

A) a > b > c > d **B)** a > c > d > b **C)** c > a > b > d **D)** a > c > b > d

38. In zinc blende(ZnS) distance between two closest Zn^{+2} ions is 5 $A°$, then distance between S^{2-} ion and its 4th nearest neighbour is

A) 5 $A°$ **B)** 3.53 $A°$ **C)** 10 $A°$ **D)** 7.07 $A°$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

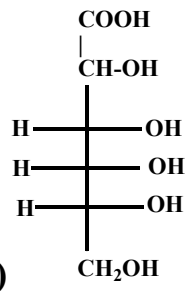
39.

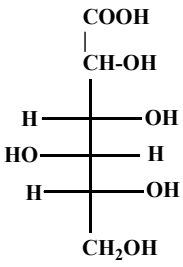
D - Aldopentose (X) $\xrightarrow{\text{con } HNO_3}$ Meso acid

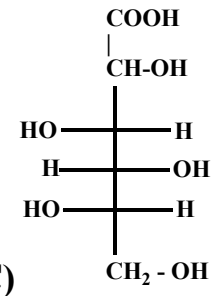
(X) $\xrightarrow[H_2O]{Br_2}$ Optical active acid

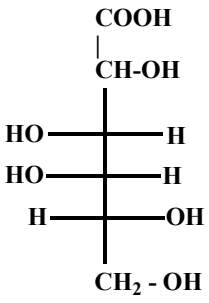
$$X \xrightarrow[NaCN]{HCN} Y \xrightarrow{H_3O^+} Z$$

Z can be ?

A)


B)


C)


D)


40. $Pb^{+2}_{aq} \xrightarrow{dil\ H_2SO_4} A \xrightarrow[CH_3COOH/\Delta]{CH_3COONH_4} B + C$

Both A & B contain same metal
Correct statement(s) is/are

A) B is ammonium tetracetatoplumbate (II)

B) A is white ppt which is insoluble in dil HCl but soluble in hot con HCl

C) C can give white ppt with $BaCl_2$ solution

D) both A & B are white ppt

41. A first order reaction: $A_{(g)} \rightarrow 3B_{(g)}$ is performed in a cylinder piston arrangement, at constant temperature and pressure, starting with 2 moles $A_{(g)}$ and 1 mole $B_{(g)}$. After 90 minutes, from start of reaction, the molar ratio of $A_{(g)}$ to $B_{(g)}$ becomes 1:25. The initial volume of system was 12L. The information (s) related is/are:

A) $t_{1/2}$ of reaction = 30 min **B)** Volume of system at t = 90 min is 26L

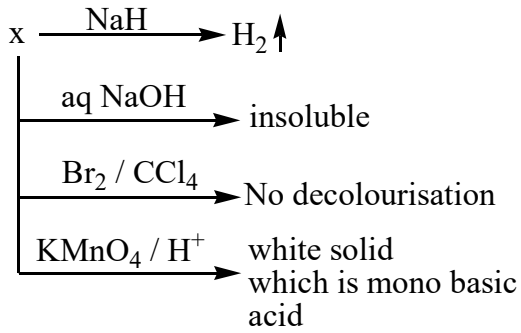
C) At t = 90 min, [A] = $\frac{1}{48}$ M **D)** At t = 90 min, [B] = $\frac{25}{104}$ M`

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

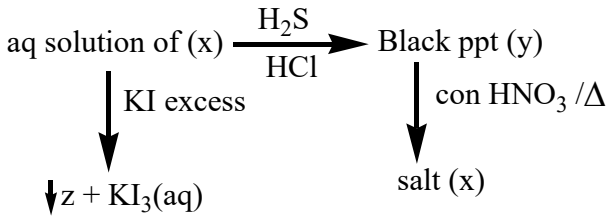
42. Optically active compounds (x) having molecular formula $C_9H_{12}O$ show following reactions



Then how many isomeric 'x' satisfy above reaction conditions.

43. $salt(x) \xrightarrow[\Delta]{con\ H_2SO_4} Brown\ gas(y)$

i) liberation of gas Y is intensified by addition of copper turnings



ii)

If P = total number of atoms in (x) and Q = oxidation state on metal in z then P – Q is

44. In how many of the following O_2 gas liberates

a) $NaNO_3 \xrightarrow{\Delta}$ b) $LiNO_3 \xrightarrow{\Delta}$

c) $Pb_3O_4 \xrightarrow{\Delta}$ d) $NH_4NO_2 \xrightarrow{\Delta}$

e) $KMnO_4 \xrightarrow{>513K}$ f) $KClO_3 \xrightarrow[\Delta]{MnO_2}$

g) $Li_2CO_3 \xrightarrow{\Delta}$ h) $Ag_2CO_3 \xrightarrow[heating]{strong}$

45. According to MOT if there is no S-P mixing then how many of the following transformations bond order decreases

a) $B_2 \rightarrow B_2^+$ b) $C_2 \rightarrow C_2^-$ c) $N_2 \rightarrow N_2^+$ d) $N_2^+ \rightarrow N_2^{+2}$

e) $Be_2^{-1} \rightarrow Be_2^{-2}$ f) $B_2 \rightarrow B_2^-$

46. A gas obeys the equation: P(V-nb) = nRT. At what pressure (in atm) and 27° C, the density of gas becomes 8.0 g/L ? The Molar mass of gas is 68 g/mol and b = 0.5 L/mol.

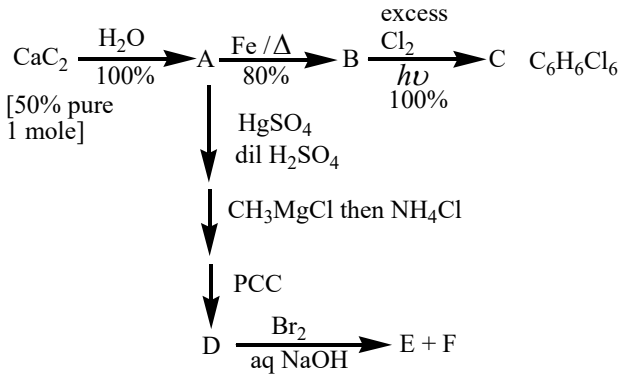
$\left(R = \frac{0.08L - atm}{mol - k} \right)$

47. Vapour pressure of A and B mixture at $80°C$ is given by P(in torr) = 100+200 X_A, X_A is mole fraction of A. Solution is prepared by mixing 2 mol of A and 3mol of B and the vapours are removed and condensed into liquid and again brought to the temperature of $80°C$, if the mole fraction of A in vapour state and mole fraction of B in vapour state above the condensate is $x:1$, find the value of 'x'.

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Paragraph-I



Consider At. Wt of H = 1, C = 12, Cl = 35.5, Br = 79.90

48. Amount of 'C' formed is _____ (gms)

49. Maximum amount Br_2 is required to get complete reaction if 5.8 gms of D is taken _____ (gms)

Paragraph-II

When $Fe_{0.96}O$ heated in the presence of oxygen, it converts to Fe_2O_3.
(At wt of Fe = 56, O=16)

50. The equivalent weight of $Fe_{0.96}O$ is _____

51. The % composition of Fe^{+2} in $Fe_{0.96}O$ is _____